SE LAB 7 – FUZZ TESTING FOR INPUT VALIDATION BUGS

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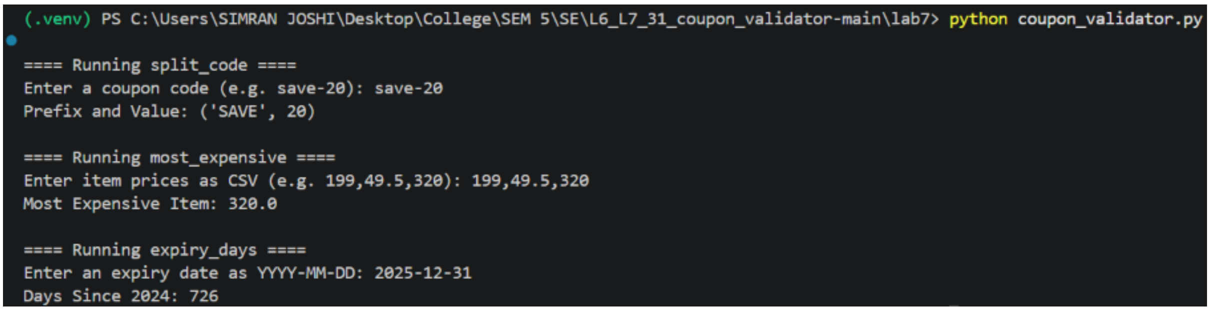
H Section

Figure 1: Original application execution



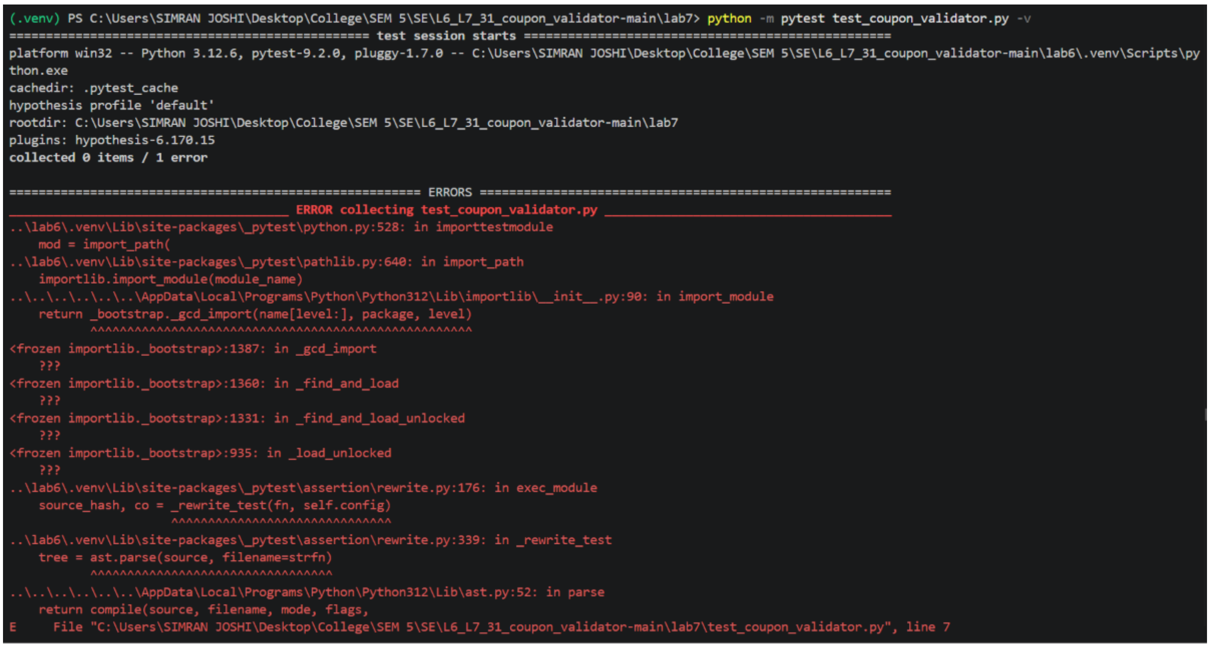
```
(.venv) PS C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7> python coupon_validator.py

==== Running split_code ====
Enter a coupon code (e.g. save-20): save-20
Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-12-31
Days Since 2024: 726
```

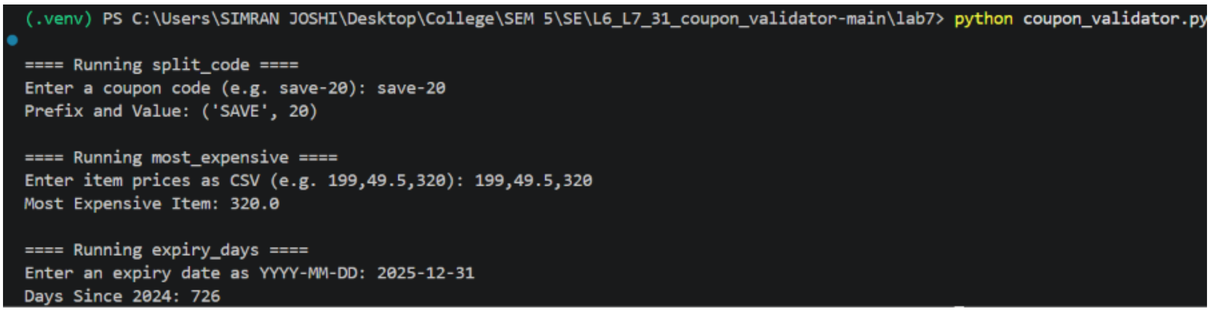
Figure 2: Hypothesis tests identifying input validation bugs



```
(.venv) PS C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7> python -m pytest test_coupon_validator.py -v
================================================= test session starts ==================================================
platform win32 -- Python 3.12.6, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab6\.venv\Scripts\py
thon.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7
plugins: hypothesis-6.170.15
collected 0 items / 1 error

======================================================== ERRORS ========================================================
_______________________________________ ERROR collecting test_coupon_validator.py _______________________________________
..\lab6\.venv\Lib\site-packages\_pytest\python.py:528: in importtestmodule
    mod = import_path(
..\lab6\.venv\Lib\site-packages\_pytest\pathlib.py:640: in import_path
    importlib.import_module(module_name)
..\..\..\..\..\..\AppData\Local\Programs\Python\Python312\Lib\importlib\__init__.py:90: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
<frozen importlib._bootstrap>:1387: in _gcd_import
    ???
<frozen importlib._bootstrap>:1360: in _find_and_load
    ???
<frozen importlib._bootstrap>:1331: in _find_and_load_unlocked
    ???
<frozen importlib._bootstrap>:935: in _load_unlocked
    ???
..\lab6\.venv\Lib\site-packages\_pytest\assertion\rewrite.py:176: in exec_module
    source_hash, co = _rewrite_test(fn, self.config)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
..\lab6\.venv\Lib\site-packages\_pytest\assertion\rewrite.py:339: in _rewrite_test
    tree = ast.parse(source, filename=strfn)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
..\..\..\..\..\..\AppData\Local\Programs\Python\Python312\Lib\ast.py:52: in parse
    return compile(source, filename, mode, flags,
E     File "C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7\test_coupon_validator.py", line 7
```

Figure 3: Execution of the fixed application



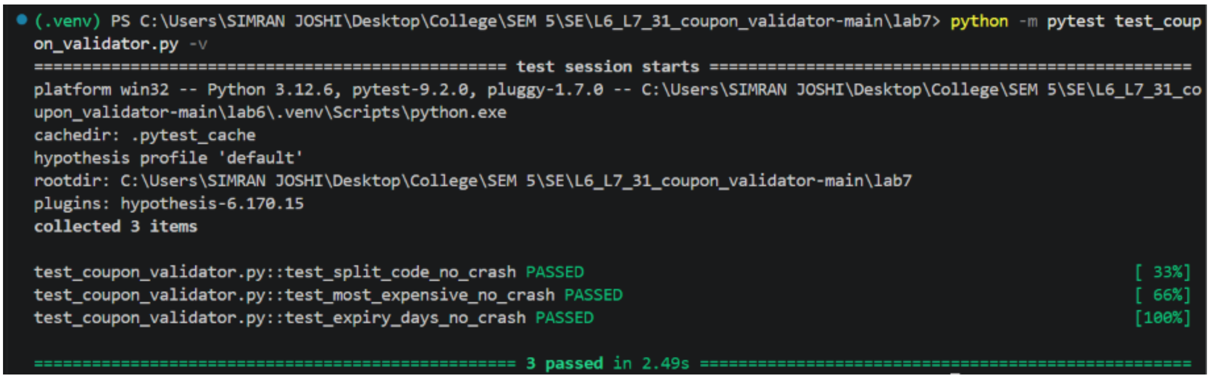
```
(.venv) PS C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7> python coupon_validator.py

==== Running split_code ====
Enter a coupon code (e.g. save-20): save-20
Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-12-31
Days Since 2024: 726
```

Figure 4: Successful execution of Hypothesis tests after fixing the bugs



```
(.venv) PS C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7> python -m pytest test_coup
on_validator.py -v
================================================= test session starts ==================================================
platform win32 -- Python 3.12.6, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_co
upon_validator-main\lab6\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\SIMRAN JOSHI\Desktop\College\SEM 5\SE\L6_L7_31_coupon_validator-main\lab7
plugins: hypothesis-6.170.15
collected 3 items

test_coupon_validator.py::test_split_code_no_crash PASSED                                                         [ 33%]
test_coupon_validator.py::test_most_expensive_no_crash PASSED                                                     [ 66%]
test_coupon_validator.py::test_expiry_days_no_crash PASSED                                                        [100%]

================================================== 3 passed in 2.49s ===================================================
```